# PATTERN RECOGNITION AND MACHINE LEARNING

## Lecture 11 – Dimensionality Reduction: Linear, Global, and Local Structure

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# Three periods, three kinds of structure

| Time | Main question | What you will do |
|---|---|---|
| Period 1: 45 min | Which linear direction keeps useful variation? | Calculate centering, covariance, eigenvectors, projection and reconstruction. |
| Period 2: 45 min | How do we compute PCA and preserve global relations? | Connect SVD to PCA; select dimensions; understand kernel PCA and calculate MDS. |
| Period 3: 45 min | What if the data lie on a curved surface? | Compare Isomap, LLE, Laplacian eigenmaps, t-SNE and UMAP; justify an embedding. |

**Learning target**

Explain the meaning of an embedding before interpreting its picture.

# A useful representation is smaller and still meaningful

- A $28 \times 28$ image has 784 pixel features; a sensor record may contain 500 measurements. Many features move together.
- **Compression:** store and process fewer numbers while controlling reconstruction error.
- **Denoising:** remove small, unstable variations when they are mostly measurement noise.
- **Visualization:** display neighborhoods in two or three coordinates, accepting some distortion.
- **Prediction:** reduce redundant inputs, then check whether a classifier actually improves.

**Different goals**

A two-dimensional picture can look clean while discarding a feature needed for classification.

# Choose an information budget before reducing the data

| Purpose | Explicit requirement | Evidence |
|---|---|---|
| Archive compression | Mean squared error below 1 | Decode and measure error |
| Exploratory display | Retain useful local neighbors | Compare neighbor identities |
| Fault classification | Preserve rare-fault recall | Evaluate labeled validation data |

- A requirement determines which losses and dimensions deserve comparison.
- A storage limit and a prediction target can lead to different representations of the same records.
- Report the chosen requirement rather than saying that a method simply preserves “information.”

# High-dimensional neighborhoods require much more coverage

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## A volume calculation

For a uniform unit cube, an axis-aligned box of side length $a$ occupies fraction $a^d$ of the total volume.

$$a^d = 0.1 \implies a = 0.1^{1/d}.$$

| Dimension $d$ | 1 | 10 | 100 |
|---|---|---|---|
| Side length for 10% of volume | 0.100 | 0.794 | 0.977 |

- In 100 dimensions, a box containing 10% of uniform probability mass spans almost the whole range on every axis.
- Nearby samples may be scarce; distance depends strongly on scaling and irrelevant features.
- This calculation describes uniform data, not every real distribution.

# Independent binary features require exponentially many patterns

$$N_{\text{patterns}} = 2^d.$$

| Features $d$ | Possible patterns | If 1,000 rows are available |
|---|---|---|
| 10 | $1,024$ | Almost one row per pattern on average |
| 20 | $1,048,576$ | Fewer than one row per thousand patterns |

- This example assumes all binary combinations are relevant and comparably likely.
- Adding ten features multiplies the possible input patterns by $1,024$.
- Real dependence can reduce the effective variety; that is one reason a lower-dimensional representation may work.

# Irrelevant features can overwhelm a meaningful distance

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## Two observations

One task-relevant feature differs by 1. There are 100 irrelevant features, each differing by 0.2.

$$\|x_i - x_j\|^2 = 1^2 + 100(0.2)^2 = 5.$$

- The useful coordinate contributes only $1/5$ of the squared distance.
- Independent noise differences accumulate even when each one looks small.
- Centering does not remove these differences; scaling can change their influence but does not identify relevance by itself.
- Check the representation before trusting the nearest-neighbor graph. A distance can be numerically exact and scientifically unhelpful.

# Ambient dimension and intrinsic dimension are different

- **Ambient dimension** $d$: the number of recorded coordinates.
- **Intrinsic dimension**: the number of independent factors needed to move along the data.
- A point on a sphere uses three coordinates $(x, y, z)$ but has only two independent directions locally.
- A rolled sheet is curved in 3-D, yet moving along the sheet still needs two coordinates.

### A representation question

Can we describe position along the sheet without mixing points on different layers? PCA uses a flat subspace; manifold methods try to follow the surface.

### Noise

Observed data only approximate an ideal low-dimensional surface.

# A nonlinear parameter can describe a low-dimensional surface

$$x(\theta, \varphi) = \big(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta\big).$$

- Three recorded coordinates describe a point on the unit sphere.
- Two parameters move locally along the surface; $x_1^2 + x_2^2 + x_3^2 = 1$ constrains the coordinates.
- At $\theta = \pi/2$, changing $\varphi$ moves around the equator: $(1, 0, 0)$ to $(0, 1, 0)$.
- This low-dimensional description is nonlinear and needs local coordinate care near the poles.
- Intrinsic dimension concerns independent movement, not simply the number of nonzero columns.

# State what the embedding should preserve

| Method family | Quantity to preserve | Question after reduction |
|---|---|---|
| PCA | Variance in an orthogonal linear subspace | How much squared reconstruction error remains? |
| MDS | Pairwise distances | Are original and embedded distances close? |
| Isomap | Distances along a neighborhood graph | Does the graph follow the surface? |
| LLE / graph methods | Local reconstruction or neighboring relations | Are local relations still meaningful? |
| t-SNE / UMAP | Local similarity structure | Are important neighbors retained? |

**Common limitation**

No map into two dimensions can preserve every relation in arbitrary high-dimensional data.

# Center four observations before choosing a direction

## One dataset for the whole PCA calculation

Rows are observations; columns are features.

$$X = \begin{bmatrix} 4 & 3 \\ 3 & 4 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \bar{x} = (2, 2), \qquad X_c = X - \mathbf{1}\bar{x}^\top = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{bmatrix}.$$

- PCA describes variation around the mean, not distance from the original origin.
- Store the training mean: a future observation must use the same center.
- These points vary together along a diagonal direction; this dataset will continue through covariance, projection and reconstruction.

# Use the same training mean for a future observation

**Continue the four-row dataset**

Training mean is $\bar{x} = (2, 2)$. Two future observations are $x_A = (4, 2)$ and $x_B = (0, 3)$.

$$x_{c,A} = (2, 0), \qquad x_{c,B} = (-2, 1).$$

- If the single new observation defines its own mean, it becomes the zero vector and loses its position.
- The future batch mean $(2, 2.5)$ would instead produce $(2, -0.5)$ and $(-2, 0.5)$, shifting the fitted origin.
- Store the fitted mean. Transforming future inputs is different from refitting a representation.

# Covariance measures how features vary together

$$X_c = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{bmatrix}, \qquad S = \frac{1}{n} X_c^\top X_c = \begin{bmatrix} 2.5 & 2 \\ 2 & 2.5 \end{bmatrix}, \quad n = 4.$$

- $S_{11} = (4 + 1 + 1 + 4)/4 = 2.5$: variance of feature 1.
- $S_{12} = (2 + 2 + 2 + 2)/4 = 2$: positive co-variation of the two features.
- $S$ is $d \times d$, symmetric and positive semidefinite; $X_c$ is $n \times d$.
- Using $1/(n-1)$ rescales every eigenvalue but leaves principal directions unchanged.

# Read every entry of the covariance matrix

$$X_c^\top X_c = \begin{bmatrix} 10 & 8 \\ 8 & 10 \end{bmatrix}, \qquad S = \tfrac{1}{4} X_c^\top X_c.$$

- Diagonal entry: $2^2 + 1^2 + (-1)^2 + (-2)^2 = 10$.
- Cross entry: $(2)(1) + (1)(2) + (-1)(-2) + (-2)(-1) = 8$.
- Division by four gives variances 2.5 and covariance 2.
- The off-diagonal entry is not a distance or a class score; it summarizes coordinated movement of the two centered features.
- Positive diagonal entries alone do not identify the principal direction. The cross terms matter.

# Scaling changes the question that PCA answers

| Representation | What dominates variance? | Appropriate when |
|---|---|---|
| Centered raw features | Large physical variation and large units | Units are comparable and variance has a physical meaning. |
| Centered z-scores | Correlation after unit removal | Different units are arbitrary relative to the task. |

**Unit conversion**

Temperature variance is 4 in degrees Celsius; pressure variance is 10,000 in Pa. Converting Pa to kPa divides its variance by $10^6$. Raw PCA changes although the observations describe the same physics.

**Decision**

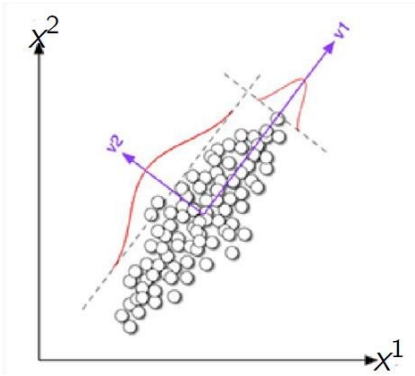
Standardization is a modeling choice, not an automatic improvement.

# Correlation PCA rescales covariance before finding directions

$$s_1 = s_2 = \sqrt{2.5}, \qquad R = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}.$$

- $R_{12} = 2/(\sqrt{2.5}\sqrt{2.5}) = 0.8$ is the correlation.
- Eigenvalues are 1.8 and 0.2; the directions remain $(1,1)/\sqrt{2}$ and $(1,-1)/\sqrt{2}$ here.
- The retained fraction is still $1.8/2 = 90\%$ because both original feature scales were equal.
- Unequal rescaling can change the directions. This special example should not be generalized to all datasets.
- Keep reconstruction units consistent with the fitted scaler.

# A principal-component score is a position along an axis




$$z = w^\top x_c, \qquad \|w\| = 1.$$

- $w$ is a direction; $z$ is a scalar coordinate, not a class label.
- For $w = (1,1)/\sqrt{2}$ and $x_c = (2,1)$, $z = 3/\sqrt{2} \approx 2.121$.
- The reconstruction is $\hat{x}_c = zw = (1.5, 1.5)$.
- Replacing $w$ by $-w$ flips all scores, but produces the same reconstructed point.

# The second score is the variation discarded by one-component PCA

$$z_1 = \frac{x_{c,1} + x_{c,2}}{\sqrt{2}}, \qquad z_2 = \frac{x_{c,1} - x_{c,2}}{\sqrt{2}}.$$

| Observation | $z_1$ | $z_2$ |
|---|---|---|
| $(2, 1)$ | $3/\sqrt{2}$ | $1/\sqrt{2}$ |
| $(1, 2)$ | $3/\sqrt{2}$ | $-1/\sqrt{2}$ |

- Both observations share their first score but differ in the second.
- Keeping both orthonormal scores is a change of coordinates, with no information loss.
- Discarding the second score merges these two positions in the one-dimensional representation.

# PCA chooses the direction with the largest projected variance

$$\mathrm{Var}(z) = \frac{1}{n}\sum_i (w^\top x_{c,i})^2 = w^\top S w, \qquad \max_{\|w\|=1} w^\top S w.$$

## Compare three directions

For $S = \begin{bmatrix} 2.5 & 2 \\ 2 & 2.5 \end{bmatrix}$:

| Direction $w$ | Projected variance | Interpretation |
|---|---|---|
| $(1, 0)$ | 2.5 | Original first feature |
| $(1, 1)/\sqrt{2}$ | 4.5 | Features increase together |
| $(1, -1)/\sqrt{2}$ | 0.5 | Features change oppositely |

## Meaning

The first direction captures the largest spread of centered observations.

# Calculate projected variance from the actual scores

$$z_1 = \left(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}, -\frac{3}{\sqrt{2}}, -\frac{3}{\sqrt{2}}\right)^\top.$$

$$\bar{z}_1 = 0, \qquad \frac{1}{4}\sum_i z_{1,i}^2 = \frac{1}{4}\left(\frac{9}{2} + \frac{9}{2} + \frac{9}{2} + \frac{9}{2}\right) = 4.5.$$

- Along the perpendicular axis, scores are $\pm 1/\sqrt{2}$, giving variance 0.5.
- The two score variances add to the original total variance 5.
- This direct calculation checks the eigenvalue interpretation without relying on a scatter plot.
- A component score belongs to an observation; a component direction belongs to the feature space.

# The variance objective leads to an eigenvector

$$\mathcal{L}(w, \lambda) = w^\top S w - \lambda(w^\top w - 1), \quad \nabla_w \mathcal{L} = 2Sw - 2\lambda w = 0.$$
$$Sw = \lambda w, \qquad w^\top S w = \lambda \quad (\|w\| = 1).$$

- $w$: a direction unchanged by multiplication with covariance, except for scale.
- $\lambda$: variance along that direction.
- Choose the eigenvector with the largest eigenvalue for the first component.
- For $m$ components, choose $m$ orthonormal eigenvectors with the largest eigenvalues.

**Scope**

This derivation optimizes squared linear projection error; it does not use class labels.

# Verify an eigenpair and the orthogonality condition

$$\begin{bmatrix} 2.5 & 2 \\ 2 & 2.5 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 4.5 \\ 4.5 \end{bmatrix} = 4.5 w_1.$$

$$w_1^\top w_2 = \tfrac{1}{2}(1-1) = 0, \qquad \|w_1\|^2 = 1.$$

- The first calculation verifies direction and eigenvalue together.
- Orthogonality prevents selecting the same direction twice when several components are kept.
- Unit length prevents artificial variance increases from enlarging the direction vector.
- Check both constraints; a vector can be orthogonal to another vector without having unit length.

# Solve the two-dimensional PCA example

$$\det(S - \lambda I) = (2.5 - \lambda)^2 - 4 = 0 \quad \Rightarrow \quad \lambda_1 = 4.5,\ \lambda_2 = 0.5.$$

$$w_1 = \frac{1}{\sqrt{2}}(1, 1)^\top, \qquad w_2 = \frac{1}{\sqrt{2}}(1, -1)^\top.$$

- Total variance is $\mathrm{tr}(S) = 5 = 4.5 + 0.5$.
- One component retains $4.5/5 = 90\%$ of the total variance.
- Check directly: $Sw_1 = 4.5w_1$ and $w_1^\top w_2 = 0$.
- Equal eigenvalues would allow multiple equally good choices of principal axes.

# Equal eigenvalues mean that the principal axes are not unique

$$S = 2I = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \qquad w^\top S w = 2 \quad \text{for every unit vector } w.$$

- Horizontal, vertical and diagonal axes all retain the same one-dimensional variance.
- PCA does not identify a preferred direction inside this isotropic subspace.
- If eigenvalues are nearly equal, small data perturbations can rotate individual axes considerably.
- The retained subspace may still be stable even when individual axis directions are not.
- Compare reconstruction and subspace relationships rather than demanding identical signs or axes from different implementations.

# Projection and reconstruction complete the calculation

| $x_c$ | $z = x_c^\top w_1$ | $\hat{x}_c = z w_1$ | $\|x_c - \hat{x}_c\|^2$ |
|---|---|---|---|
| $(2, 1)$ | $3/\sqrt{2}$ | $(1.5, 1.5)$ | 0.5 |
| $(1, 2)$ | $3/\sqrt{2}$ | $(1.5, 1.5)$ | 0.5 |
| $(-1, -2)$ | $-3/\sqrt{2}$ | $(-1.5, -1.5)$ | 0.5 |
| $(-2, -1)$ | $-3/\sqrt{2}$ | $(-1.5, -1.5)$ | 0.5 |

$$Z = X_c W_m, \qquad \hat{X} = Z W_m^\top + \mathbf{1}\bar{x}^\top.$$

- $W_m$ is $d \times m$; $Z$ is $n \times m$; reconstruction returns to $n \times d$.
- Add the original training mean only when decoding back to original coordinates.

# Decode back to the original units and mean

**Return to raw observation** $(4, 3)$

Its centered version is $(2, 1)$ and its one-component reconstruction is $(1.5, 1.5)$.

$$\hat{x} = (1.5, 1.5) + (2, 2) = (3.5, 3.5), \qquad x - \hat{x} = (0.5, -0.5).$$

- The residual squared length is $0.25 + 0.25 = 0.5$ in the original centered scale.
- Adding the mean changes location but not the reconstruction error.
- With a fitted scaler, multiply reconstructed standardized coordinates by their stored scales before adding means.
- Decoding is part of evaluating compression, not merely another plotting operation.

# Maximum variance and minimum reconstruction error agree

$$\|x_c\|^2 = \|W_m^\top x_c\|^2 + \|x_c - W_m W_m^\top x_c\|^2.$$

$$\frac{1}{n}\|X_c - \hat{X}_c\|_F^2 = \mathrm{tr}(S) - \sum_{j=1}^{m} \lambda_j = \sum_{j=m+1}^{d} \lambda_j.$$

- Orthogonal projection splits squared length into retained and discarded parts.
- Total centered variation is fixed, so retaining more variance leaves less reconstruction error.
- Our example: $5 - 4.5 = 0.5$ per observation when one component is retained.
- The equivalence uses orthonormal axes and squared Euclidean error.

# A reconstruction budget determines how many components are needed

### Three eigenvalues

Let $\lambda = (4, 1, 0.1)$. Mean squared reconstruction error is the sum of discarded eigenvalues.

| Retained $m$ | Error $E(m)$ | Meet error budget 0.2? |
|---|---|---|
| 1 | $1 + 0.1 = 1.1$ | No |
| 2 | 0.1 | Yes |
| 3 | 0 | Yes, but stores more coordinates |

- Select two components if the goal is the smallest representation meeting this budget.
- A label-dependent task may impose another requirement beyond reconstruction.
- The budget must use the same scaling and error convention as the eigenvalues.

# Exercise: interpret PCA without relying on a picture

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**Work in pairs, then explain your answer**

A centered dataset has covariance $S = \mathrm{diag}(9, 1, 0.25)$.

1. If one component is kept, what is the principal direction?
2. What fraction of total variance is retained?
3. What is the average squared reconstruction error?
4. A class label depends only on feature 3. Is one-component PCA necessarily a good classifier preprocessing step?

**After discussion**

Direction $(1, 0, 0)$; variance fraction $9/10.25 \approx 87.8\%$; error 1.25. The label-related feature may be discarded.

# Solution workshop: separate direction, retained variance and error

$$S = \mathrm{diag}(9, 1, 0.25), \qquad \mathrm{tr}(S) = 10.25.$$

- One component selects the first coordinate; retained fraction is $9/10.25 \approx 87.8\%$.
- The discarded average squared error is $1 + 0.25 = 1.25$.
- Two components retain $10/10.25 \approx 97.6\%$ and leave error $0.25$.
- If labels depend on coordinate 3, neither unsupervised choice guarantees good classification.
- Explain which number answers which question. Variance percentage, reconstruction error and classification accuracy have different denominators and meanings.

# SVD computes PCA directly from centered observations

$$X_c = U\Sigma V^\top, \qquad S = \frac{1}{n} V\Sigma^\top \Sigma V^\top.$$

- Rows of $X_c \in \mathbb{R}^{n\times d}$ are observations. Columns of $V$ are feature-space principal directions.
- Singular values satisfy $\lambda_j = \sigma_j^2/n$ under our covariance convention.
- Scores are $Z = X_c V_m = U_m \Sigma_m$.
- Forming $X_c^\top X_c$ squares the condition number. Direct SVD can be more numerically reliable.
- At most $\min(n-1, d)$ centered components have nonzero variance.

# Thin SVD has separate sample and feature coordinates

$$X_c \in \mathbb{R}^{n\times d}, \quad U_r \in \mathbb{R}^{n\times r}, \quad \Sigma_r \in \mathbb{R}^{r\times r}, \quad V_r \in \mathbb{R}^{d\times r}.$$
$$X_c = U_r \Sigma_r V_r^\top, \qquad r \le \min(n-1, d).$$

- $V_r$ contains directions in feature space; $U_r\Sigma_r$ contains scores for the observed rows.
- With 20 centered observations and 500 features, $r \le 19$; a $500 \times 500$ covariance need not be explicitly formed.
- The Chinese reference uses column observations, so its feature directions are left singular vectors. Both conventions are valid when used consistently.

# The SVD numbers must agree with the PCA numbers

## Use the same four observations

For $n = 4$, covariance eigenvalues are 4.5 and 0.5.

$$\sigma_1 = \sqrt{n\lambda_1} = \sqrt{18} \approx 4.243, \qquad \sigma_2 = \sqrt{2} \approx 1.414.$$

| Quantity | Component 1 | Component 2 |
|---|---|---|
| Variance $\sigma_j^2/n$ | 4.5 | 0.5 |
| Variance fraction | 0.90 | 0.10 |
| Feature direction | $(1,1)/\sqrt{2}$ | $(1,-1)/\sqrt{2}$ |

## Sanity check

$\|X_c\|_F^2 = \sigma_1^2 + \sigma_2^2 = 20$; dividing by four gives total variance five.

# Recover the first left singular vector from the PCA scores

$$z_1 = (3, 3, -3, -3)^\top / \sqrt{2}, \qquad \sigma_1 = \sqrt{18}.$$
$$u_1 = z_1 / \sigma_1 = (1, 1, -1, -1)^\top / 2, \qquad \|u_1\|^2 = 1.$$

- Multiplying $u_1$ by $\sigma_1$ returns the four observed component scores.
- Multiplying $u_1 \sigma_1 w_1^\top$ reconstructs the rank-one centered matrix.
- Its rows are $(1.5, 1.5), (1.5, 1.5), (-1.5, -1.5), (-1.5, -1.5)$.
- The omitted singular value contributes squared matrix error $\sigma_2^2 = 2$, or 0.5 per observation.

# Choose the dimension using a stated requirement

$$R(m) = \frac{\sum_{j=1}^{m} \lambda_j}{\sum_{j=1}^{d} \lambda_j}, \qquad E(m) = \sum_{j=m+1}^{d} \lambda_j.$$

**Eigenvalues** $(6, 3, 1)$

One component retains 60% with error 4; two retain 90% with error 1; three retain everything with error 0.

- Compression: choose the smallest $m$ meeting a variance or reconstruction requirement.
- Prediction: compare candidate $m$ values using validation performance inside a fitted pipeline.
- Visualization: two dimensions are a display constraint, not proof that the data are intrinsically 2-D.
- A large spectral gap suggests a dimension; it does not replace task-based evidence.

# Cumulative variance makes the dimension choice explicit

**Spectrum** $(6, 3, 0.7, 0.3)$

Total variance is 10.

| Keep $m$ | Cumulative variance | Retained fraction | Error |
|---|---|---|---|
| 1 | 6 | 60% | 4 |
| 2 | 9 | 90% | 1 |
| 3 | 9.7 | 97% | 0.3 |
| 4 | 10 | 100% | 0 |

- A 95% variance requirement selects three components; a 90% requirement selects two.
- A visible spectral gap is a suggestion, not an independent task guarantee.
- State whether your criterion concerns compression, noise or prediction.

# Whitening removes variance scale and may amplify small components

$$\tilde{z}_j = z_j/\sqrt{\lambda_j}, \qquad \mathrm{Var}(\tilde{z}_j) = 1 \quad (\lambda_j > 0).$$

## Our PCA example

The first score $3/\sqrt{2}$ is divided by $\sqrt{4.5}$ to become 1; the second score $1/\sqrt{2}$ divided by $\sqrt{0.5}$ also becomes 1.

- Whitening keeps axes but removes their relative variance scale.
- Small eigenvalues receive stronger amplification; nearly zero noise directions may become unstable.
- Dimensional truncation and whitening are separate decisions.
- Use whitening only when equalized component scales suit the downstream task.

# A small-variance feature can determine the class

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## Two independent features

Feature 1 is nuisance variation with variance 100. Feature 2 is a small signal: class means are $-0.1$ and $+0.1$, with within-class standard deviation 0.01.

- Raw one-component PCA selects feature 1 because its variance is much larger.
- The selected coordinate contains no class information if the nuisance is independent of the label.
- Feature 2 separates the class means by 20 within-class standard deviations.
- Feature selection, supervised reduction or a validation-tested representation may preserve the useful signal.

## Lesson

High explained variance answers a reconstruction question. Classification requires another check.

# Contrast PCA with the supervised separation objective

| Method | Information used | Direction favored |
|---|---|---|
| PCA | Input covariance | Large total variance |
| Fisher linear discriminant | Labels, class means and within-class scatter | Large between-class separation relative to within-class variation |

- A high-variance nuisance may dominate PCA while contributing nothing to class separation.
- A low-variance signal can be valuable if its class means are well separated relative to noise.
- This connects the present unsupervised lecture with the labeled discriminant methods from Lecture 4.
- Compare the complete task pipeline rather than treating either objective as universally preferable.

# Fit preprocessing and PCA inside the training split

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1. Split labeled observations into training, validation and test sets before fitting any transformation.
2. Fit the scaler and PCA on training rows. Store means, scales and principal directions.
3. Transform validation rows with these stored quantities; compare dimensions and downstream models.
4. Refit the selected pipeline on training plus validation only after choices are fixed.
5. Evaluate the final pipeline once on untouched test data.

## A future observation

A new $x$ uses $z = W_m^\top [(x - \bar{x})/s]$ if standardization was part of the training pipeline. It does not define a new mean or new axes.

# A fitted PCA pipeline keeps validation coordinates comparable

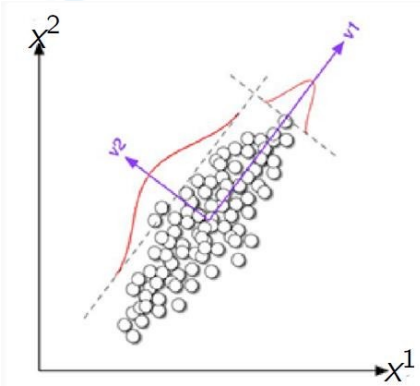
1. Fit scaler means and scales using training rows only.
2. Fit PCA directions on those transformed training rows.
3. Apply the stored scaler and PCA to validation rows.
4. Fit and select the downstream classifier using a consistent validation protocol.

## Tiny check

If training mean is 2 and scale is 2, a validation value 6 becomes $(6 - 2)/2 = 2$. It does not become zero by centering on itself.

- Cross-validation repeats fitting inside each training fold.
- Saving a classifier without its scaler and PCA does not save the complete predictor.

# Read the principal axes rather than just the scatter plot




- The long direction corresponds to the larger eigenvalue.
- The short perpendicular direction captures residual variation.
- Projecting onto the long axis keeps the largest possible linear variance for one coordinate.
- Reconstructing moves each point back onto that axis; the perpendicular gap is the error.
- The axis sign is arbitrary; axis length and retained variance have precise meanings.

**Explain aloud**

PCA rotates to orthogonal axes and then discards selected coordinates.

# A flat subspace cannot always unfold a curved structure

- PCA maps $x$ to $W_m^\top x_c$: every output coordinate is a linear combination of the original features.
- A ring projected onto one line maps different positions to the same coordinate.
- A rolled sheet projected onto a plane may mix points from different layers.
- A nonlinear feature map $\phi(x)$ may make the structure simpler before a linear projection.
- Kernel PCA performs PCA in an implicit feature space; Isomap and local methods use other structural assumptions.

**Do not overpromise**

A nonlinear method can reveal useful structure, but its geometry depends on its kernel or neighborhood choices.

# A nonlinear feature can reveal structure that a line cannot

## Two concentric rings

Points from an inner ring satisfy $x_1^2 + x_2^2 \approx 1$; an outer ring has $x_1^2 + x_2^2 \approx 4$.

$$r^2(x) = x_1^2 + x_2^2.$$

- Projecting onto $x_1$ merges different angles and may mix rings.
- The nonlinear radial feature maps the two rings near 1 and 4.
- This example illustrates feature design, not an automatic guarantee that kernel PCA recovers the desired radial coordinate.
- The chosen kernel determines which nonlinear relationships become available.
- A useful nonlinear coordinate must still be matched to the actual task.

# Kernel PCA reuses the kernel trick from SVM

$$K_{ij} = k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle.$$

- Perform linear PCA on centered $\phi(x)$ without constructing every feature-space coordinate.
- For a polynomial kernel, the implicit space includes interactions; an RBF kernel describes similarity through distance.
- The matrix is $n \times n$, so memory grows with the number of samples rather than only the original dimension.
- Nonlinear coordinates depend on the kernel and its parameters; the inverse map to original inputs is generally not immediate.

**Connection**

SVM uses a kernel to fit a labeled decision boundary. Kernel PCA uses a kernel to describe input variation, without labels.

# Calculate one polynomial-kernel entry before using the matrix

$$k(x, x') = (1 + x^\top x')^2.$$

**Inputs $x = (1, 2)$ and $x' = (2, 1)$**

The dot product is $1 \times 2 + 2 \times 1 = 4$, so $k(x, x') = 25$.

- A feature map is $\phi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, \sqrt{2}x_1x_2, x_2^2)$.
- Its inner product expands to the same kernel value.
- Interactions such as $x_1x_2$ become ordinary coordinates in the implicit space.
- Kernel PCA uses all training-pair kernel entries, then centers the implicit coordinates before finding directions.
- A kernel value is an inner product, not generally an original-space distance.

# Center the kernel matrix before eigendecomposition

$$H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top, \qquad K_c = HKH.$$

$$(K_c)_{ij} = K_{ij} - \bar{K}_{i\cdot} - \bar{K}_{\cdot j} + \bar{K}_{\cdots}.$$

- $H$ subtracts the mean in the implicit feature space.
- For normalized eigenvector $v_j$ with $K_c v_j = \eta_j v_j$ and $\eta_j > 0$, the training score vector is $\sqrt{\eta_j}\, v_j$.
- Zero eigenvalues carry no variation; a valid positive semidefinite kernel has no substantial negative eigenvalues.
- Keep the centering quantities and eigenvectors for transforming future samples.

# Worked kernel centering with three scalar observations

**Linear kernel for $x = (0, 1, 2)$**

The uncentered Gram matrix is $K = xx^\top$.

$$K = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 4 \end{bmatrix}, \qquad K_c = HKH = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

- Centered coordinates are $(-1, 0, 1)$, whose pairwise products give $K_c$ directly.
- Example: $K_{c,13} = 0 - 0 - 2 + 1 = -1$ using row mean, column mean and grand mean.
- Negative off-diagonal values are valid; positive semidefinite does not mean every entry is positive.
- The largest eigenvalue is 2 and the centered training scores are $(-1, 0, 1)$ up to sign.

# A new point must use training-set kernel centering

$$k_{c,i}(x) = k(x_i, x) - \frac{1}{n}\sum_l k(x_l, x) - \frac{1}{n}\sum_l K_{il} + \frac{1}{n^2}\sum_{l,r} K_{lr}.$$

$$z_j(x) = \frac{v_j^\top k_c(x)}{\sqrt{\eta_j}}.$$

- $x_i$, row means, total kernel mean and eigenvectors all come from training data.
- Smaller RBF bandwidth emphasizes very local similarities; overly large bandwidth makes many pairs look alike.
- Compare the kernel and dimension using a task or distortion measure, not only a visually separated plot.
- Large $n$ may require approximations; kernel PCA is not automatically cheaper than linear PCA.

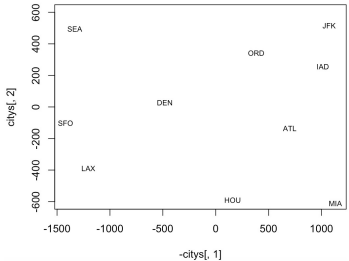
# Kernel storage grows with the number of training observations

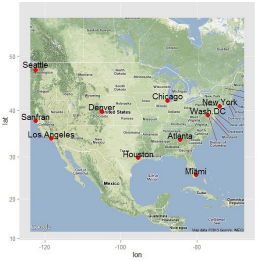
$$K \in \mathbb{R}^{n \times n}, \qquad \text{float64 storage} \approx 8n^2 \text{ bytes.}$$

| Training rows | Kernel entries | Approximate storage |
|---|---|---|
| 1,000 | One million | 8 MB |
| 10,000 | One hundred million | 800 MB |

- These figures exclude eigensolver workspaces and other arrays.
- A future point needs its kernel relation to training rows and their saved centering statistics.
- Increasing sample count tenfold multiplies dense kernel storage by one hundred.
- Approximate kernels or linear PCA may be appropriate when scale and task support them.

# A map can be reconstructed from distances alone




- Input distances in the source table's units: ATL–ORD 587; LAX–SFO 347; JFK–IAD 205.
- The recovered coordinates on the left should approximate these distances; the map on the right provides geographical context.
- Translation, rotation and reflection do not change pairwise distances. Real travel distances need not fit an exact Euclidean map.

# Distances identify relative geometry, not absolute orientation

$$d_{ij} = \|z_i - z_j\| = \|Qz_i + t - (Qz_j + t)\| \quad (Q^\top Q = I).$$

- Translation $t$ cancels from every pairwise difference.
- Orthogonal $Q$ can rotate or reflect all coordinates without changing distances.
- Scaling by two does change every distance by two, so scale is not arbitrary when numerical distances are specified.
- The city map may be rotated for geographical comparison, but orientation is extra context.
- MDS can recover a relative map while leaving these harmless coordinate choices unresolved.

# Classical MDS turns squared distances into inner products

$$D_{ij}^{(2)} = d_{ij}^2, \qquad H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top, \qquad B = -\tfrac{1}{2} H D^{(2)} H.$$

- $D^{(2)}$: squared distances, not ordinary distances.
- $B$: centered Gram matrix, with $B_{ij} = z_i^\top z_j$ if distances are Euclidean.
- Eigendecompose $B = V \Lambda V^\top$; coordinates are $Z = V_m \Lambda_m^{1/2}$.
- Retain the largest positive eigenvalues. Large negative eigenvalues indicate that the distances are not exactly Euclidean.

## Connection

PCA starts from centered coordinates; classical MDS starts from distances and recovers centered coordinates.

# For Euclidean data, classical MDS and PCA recover the same low-rank geometry

$$B = X_c X_c^\top = U\Sigma^2 U^\top, \qquad Z_{\mathrm{MDS}} = U_m \Sigma_m = Z_{\mathrm{PCA}}.$$

- PCA starts from centered feature coordinates; MDS starts from their Euclidean pairwise distances.
- Double centering reconstructs the same sample Gram matrix.
- Both then retain the largest positive spectral directions.
- Signs and rotations within equally valued subspaces can differ without changing geometry.
- Non-Euclidean distances or a directly optimized stress objective can break this equivalence.

# Double centering removes unknown squared lengths

$$d_{ij}^2 = \|z_i\|^2 + \|z_j\|^2 - 2z_i^\top z_j, \qquad \sum_i z_i = 0.$$

$$B_{ij} = -\tfrac{1}{2}\left(d_{ij}^2 - \overline{d_{i\cdot}^2} - \overline{d_{\cdot j}^2} + \overline{d_{\cdot\cdot}^2}\right).$$

- Subtract row and column means to cancel the unknown coordinate lengths.
- Add the grand mean because it was subtracted twice.
- Multiply by $-1/2$ to isolate the inner product.
- This derivation assumes centered coordinates; distance data do not determine an absolute origin anyway.

**Next calculation**

Three locations separated by distances $1, 1, 2$ can be recovered exactly on a line.

# Worked MDS: recover three points on a line

$$D^{(2)} = \begin{bmatrix} 0 & 1 & 4 \\ 1 & 0 & 1 \\ 4 & 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

- Row means of $D^{(2)}$ are $(5/3, 2/3, 5/3)$; the grand mean is $4/3$.
- Example: $B_{13} = -\frac{1}{2}(4 - 5/3 - 5/3 + 4/3) = -1$.
- One positive eigenvalue is 2, with normalized eigenvector $(1, 0, -1)/\sqrt{2}$.
- Coordinates are $Z = \sqrt{2}\,(1, 0, -1)^\top/\sqrt{2} = (1, 0, -1)^\top$.
- Recovered distances are $1, 1, 2$; reversing the line gives an equally valid result.

# Verify the MDS eigenvalue before taking its square root

$$B = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}, \quad v = (1, 0, -1)^\top / \sqrt{2}, \quad Bv = 2v.$$

$$Z = \sqrt{2}v = (1, 0, -1)^\top, \qquad ZZ^\top = B.$$

- The eigenvector gives normalized shape; the square-root eigenvalue restores coordinate scale.
- Using $2v$ instead produces distances $\sqrt{2}$ times too large.
- Recompute $|z_1 - z_2| = 1$ and $|z_1 - z_3| = 2$ to check the original distance input.
- Reconstructing the Gram matrix provides a second independent check.

# Distance stress and classical MDS are related but distinct

$$\text{Stress}(Z) = \sum_{i<j} \left(d_{ij} - \|z_i - z_j\|\right)^2.$$

- Metric MDS directly minimizes a distance-stress objective, usually by iterative optimization.
- Classical MDS uses an eigendecomposition of the double-centered squared-distance matrix; it approximates the Gram matrix.
- Both seek a useful distance-based map, but they need not give identical coordinates when distortion is unavoidable.
- Compare residual distances, inspect influential pairs and state the target dimension.
- A low numerical error does not rescue a distance measure that is irrelevant to the task.

# Some distance tables cannot be represented exactly in Euclidean space

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## Three requested distances

Suppose $d(A, B) = 1$, $d(B, C) = 1$ and $d(A, C) = 3$.

$$d(A, C) \leq d(A, B) + d(B, C) = 2 \quad \text{for Euclidean distances.}$$

- The requested value 3 violates the triangle inequality; no exact Euclidean coordinates can satisfy all three.
- An optimizer must compromise, even with an ideal numerical implementation.
- Negative eigenvalues of the double-centered matrix indicate another form of Euclidean incompatibility.
- Report residuals or a clearly qualified approximation instead of claiming that every table is an exact map.
- Check whether the input is truly a distance rather than an arbitrary dissimilarity.

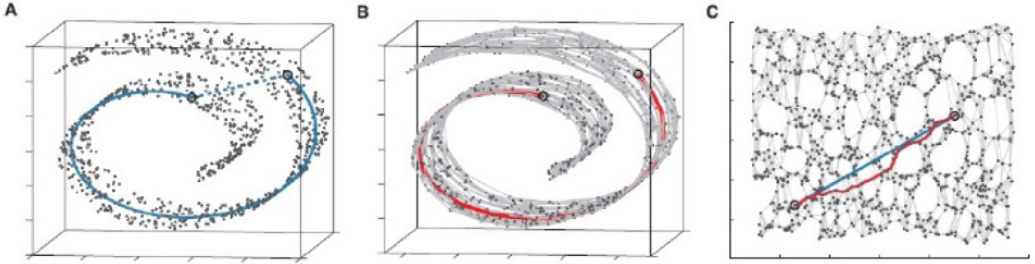
# Exercise: choose coordinates from what you are given

1. You have 20 samples with 500 features. Why can centered PCA have at most 19 nonzero components?
2. You have only a distance matrix. Which matrix must be squared and double-centered before classical MDS?
3. A one-component PCA retains 95% of input variance. What additional evidence is needed before using it for classification?
4. An MDS solution is reflected left-to-right. Are its pairwise distances changed?

**Discuss the reasons**

Centering limits rank; square the distances; evaluate a labeled validation task; reflection preserves all distances.

# On a rolled sheet, a shortcut can connect the wrong neighbors




- Panel A compares a straight shortcut with a path along the rolled surface.
- Panel B uses short local graph edges to approximate travel along the surface.
- Panel C unfolds the surface: two points may be close in 3-D but far along the sheet.

**Geodesic distance**

Distance measured along the data manifold rather than through the surrounding space.

# Local distance approximates curved distance only at a small scale

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## Unit circle

Two points separated by angle $\theta$ have arc distance $\theta$ and chord distance $2\sin(\theta/2)$.

| Angle | Arc distance | Chord distance |
|---|---|---|
| 0.2 radians | 0.200 | approximately 0.200 |
| $\pi$ radians | approximately 3.142 | 2.000 |

- For close points, the straight chord is a good local approximation.
- Across a large bend, a chord becomes a shortcut through surrounding space.
- Isomap assembles many short distances into a longer path; the approximation requires adequate sampling and appropriate edges.

# Isomap: neighborhood graph, shortest paths, then MDS

1. Build a sparse neighborhood graph using $k$ nearest neighbors or a radius $\varepsilon$.
2. Weight each local edge by its original Euclidean distance.
3. Compute shortest-path distances $d_G(i, j)$ over the graph.
4. Apply classical MDS to the squared graph-distance matrix.

- Local Euclidean distances approximate local surface distances.
- Long graph paths approximate global geodesic distances only if the graph follows the manifold.
- Disconnected components have infinite cross-component distances; do not silently replace them with arbitrary finite values.

# Write the complete shortest-path matrix for a four-node chain

## Unit-weight chain A–B–C–D

Only adjacent nodes are connected.

$$D_G = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{bmatrix}.$$

- Each entry counts the total edge length along the unique available path.
- The matrix is symmetric, has zero diagonal and satisfies the triangle inequality.
- Classical MDS can recover these nodes on a line with adjacent spacing one.
- A disconnected node would have infinite cross-component distances, not a large arbitrary finite distance.
- The graph defines the distance before MDS chooses coordinates.

# Worked Isomap: one bad edge changes global distances

## A chain along a curved surface

Edges $A$–$B$, $B$–$C$, $C$–$D$ each have length 1. There is no direct $A$–$D$ edge.

$$d_G(A, D) = 1 + 1 + 1 = 3.$$

- If $A$ and $D$ happen to be near in the surrounding space, their Euclidean distance may be 0.2.
- Adding that shortcut edge changes $d_G(A, D)$ from 3 to 0.2.
- It also changes other shortest paths, not only the shortcut pair.
- MDS then faithfully embeds the wrong graph distances.

## Interpretation

A good eigensolver cannot repair a graph that mixes different layers of the manifold.

# A shortest path minimizes edge sums, not the number of edges

## Two routes from A to D

Route A–B–D has lengths 0.8 and 1.0. Route A–C–D has lengths 0.5 and 0.5.

$$d_G(A, D) = \min(0.8 + 1.0, 0.5 + 0.5) = 1.0.$$

- Both routes contain two edges, but their geometric lengths differ.
- If a direct edge A–D of length 0.2 is added, it dominates both routes.
- Weighted paths preserve local distance information; unweighted paths only count hops.
- Choose edge weights consistently with the intended manifold distance.
- A shortcut can change many matrix entries, so inspect suspicious edges before interpreting the embedding.

# Neighborhood size controls a real trade-off

| Neighborhood | Typical risk | What to inspect |
|---|---|---|
| Too small | Fragmented graph; unstable local estimates | Connected components and sensitivity to sampling |
| Too large | Shortcuts across folds; loss of locality | Suspicious cross-layer edges and path changes |
| Reasonable range | Local coverage without major shortcuts | Stability across several neighboring parameter values |

- Standardize or choose a domain-relevant metric before constructing neighbors.
- A $k$-NN graph can be symmetrized by mutual or union neighbors; the choice changes connectivity.
- Report graph construction along with the embedding, because it defines the assumed geometry.

# Mutual and union nearest-neighbor rules produce different graphs

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## Directed neighbor lists

A chooses B; B chooses C; C chooses B.

| Symmetrization | Undirected edges | Consequence |
|---|---|---|
| Union | A–B and B–C | All three nodes connect |
| Mutual | B–C only | A is isolated |

- Union accepts an edge if either node selects the other; mutual requires both selections.
- Mutual neighbors are more conservative but can disconnect sparsely sampled regions.
- The choice affects shortest paths, local weights and later label propagation.
- Report the rule rather than merely saying “a nearest-neighbor graph.”

# LLE reconstructs each point from its neighbors

$$x_i \approx \sum_{j \in \mathcal{N}_i} w_{ij} x_j, \qquad \sum_{j \in \mathcal{N}_i} w_{ij} = 1.$$

- First choose neighbors in the original space.
- Find weights that reconstruct each point from its local neighbors.
- Keep these weights fixed and seek low-dimensional coordinates satisfying the same local combinations.
- The sum-to-one constraint makes the relation invariant to translating the local neighborhood.
- LLE preserves local reconstruction relationships, not all global distances.

**Simple picture**

A point halfway between two neighbors has weights 0.5 and 0.5; the low-dimensional map should retain that relationship.

# Worked LLE: preserve a one-quarter position

## One point and two neighbors

Target $x_i = (0.5, 0)$; neighbors $x_a = (0, 0)$ and $x_b = (2, 0)$.

$$x_i = w_a x_a + w_b x_b, \qquad w_a + w_b = 1 \quad \Rightarrow \quad w_b = 0.25,\ w_a = 0.75.$$

- The target lies one-quarter of the way from neighbor $a$ to neighbor $b$.
- If the embedding has $z_a = 0$ and $z_b = 4$, preserving these weights gives $z_i = 1$.
- The distance scale changed, but the local interpolation relationship was retained.
- With noisy or nearly dependent neighbors, regularize the local least-squares calculation.

# The sum-to-one rule preserves local relations after translation

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## Target and neighbors

Target is 0.5, neighbors are 0 and 2, and weights are $(0.75, 0.25)$. Translate every coordinate by 10.

$$0.75(10) + 0.25(12) = 10.5.$$

- The translated target is also $0.5 + 10 = 10.5$.
- More generally, $\sum_j w_j(x_j + a) = \sum_j w_j x_j + a$ when $\sum_j w_j = 1$.
- If weights summed to two, the reconstruction would add $2a$ and fail to translate correctly.
- This constraint does not guarantee nonnegative weights or exact reconstruction under noise.
- LLE preserves a local affine relation, not the original origin.

# LLE weights are not necessarily probabilities

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## Extrapolation outside two neighbors

Neighbors are $x_a = 0$ and $x_b = 1$, while the target is $x_i = 2$.

$$w_a + w_b = 1, \quad 0w_a + 1w_b = 2 \quad \Rightarrow \quad w_b = 2,\ w_a = -1.$$

- The weights reconstruct the target exactly but are not a probability distribution.
- With embedded neighbors $z_a = 10, z_b = 12$, the same relation gives $z_i = -10 + 24 = 14$.
- Negative weights can arise when a point is outside its neighbors' convex hull.
- Near-dependent noisy neighborhoods can make large weights unstable; inspect or regularize the local fit.

# LLE assembles local weights into global coordinates

$$\min_Z \sum_i \left\| z_i - \sum_j w_{ij} z_j \right\|^2 = \min_Z \operatorname{tr}(Z^\top M Z), \quad M = (I - W)^\top (I - W).$$

$$\mathbf{1}^\top Z = 0, \qquad Z^\top Z = I.$$

- Without constraints, all points could collapse to the same coordinate and obtain zero error.
- Centering removes arbitrary translation; normalization prevents collapse.
- Use the eigenvectors of $M$ with the smallest nonzero eigenvalues.
- PCA selects large variance eigenvalues; LLE selects small local-reconstruction-error eigenvalues.

# Normalization prevents every embedded point from collapsing

$$W\mathbf{1} = \mathbf{1} \quad \Rightarrow \quad (I - W)\mathbf{1} = 0.$$

- If $Z = \mathbf{1}c^\top$, every row is the same point and the LLE objective is zero.
- The centering condition $\mathbf{1}^\top Z = 0$ removes an arbitrary common offset.
- The normalization $Z^\top Z = I$ prevents all coordinates becoming zero.
- In one dimension, $z = (-1, 0, 1)/\sqrt{2}$ is centered and has unit squared length.
- Small nonzero eigenvectors describe useful variation beyond the constant mode. Constraints are necessary parts of the learning problem.

# Graph smoothness connects dimensionality reduction to the next lecture

$$A_{ij} = \text{similarity of connected points}, \quad D_{ii} = \sum_j A_{ij}, \quad L = D - A.$$

$$z^\top L z = \frac{1}{2} \sum_{i,j} A_{ij}(z_i - z_j)^2.$$

- A large edge weight penalizes placing its endpoints far apart.
- $L$ is positive semidefinite for symmetric nonnegative weights.
- A constant coordinate has zero graph variation, but is not a useful embedding by itself.
- Laplacian eigenmaps use this smoothness to produce coordinates.
- In Lecture 12, the same graph energy will spread a few known labels to unlabeled nodes.

# Construct the Laplacian matrix for a three-node chain

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad D = \operatorname{diag}(1, 2, 1), \quad L = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}.$$

- Each diagonal degree is the sum of adjacent edge weights.
- $L\mathbf{1} = 0$: a constant coordinate has zero graph variation.
- For $z = (-1, 0, 1)^\top$, $Lz = (-1, 0, 1)^\top$ and $z^\top Lz = 2$.
- The same energy is the sum of two unit-edge squared differences, $1 + 1$.
- Matrix entries encode local differences, not arbitrary class labels.

# Laplacian eigenmaps retain smooth nonconstant coordinates

$$\min_z z^\top L z, \qquad z^\top D z = 1, \quad z^\top D \mathbf{1} = 0 \quad \Rightarrow \quad L z = \lambda D z.$$

**Three-node chain with unit edges**

For $z = (-1, 0, 1)^\top$, graph energy is $(-1-0)^2 + (0-1)^2 = 2$. For $z = (-1, 1, -1)^\top$, energy is $4 + 4 = 8$.

- Small nonzero generalized eigenvalues give coordinates that vary smoothly over the graph.
- The constant zero mode is excluded because it collapses all nodes.
- This objective preserves neighboring similarity, not global Euclidean distances.

# Compare a smooth generalized eigenvector with a rapidly varying one

$$Lz = \lambda Dz, \qquad D = \mathrm{diag}(1, 2, 1).$$

- For $z = (-1, 0, 1)^\top$, $Lz = z = Dz$, so $\lambda = 1$.
- For $u = (1, -1, 1)^\top$, $Lu = (2, -4, 2)^\top = 2Du$, so $\lambda = 2$.
- The constant vector has $\lambda = 0$ and is excluded from useful coordinates.
- After the same $D$-norm normalization, the smaller nonzero eigenvalue represents smoother graph variation.
- Ordinary and generalized eigenproblems differ because degree weights enter the normalization.

# t-SNE compares probabilities of being neighbors

$$p_{j|i} = \frac{\exp[-\|x_i - x_j\|^2/(2\sigma_i^2)]}{\sum_{k\neq i}\exp[-\|x_i - x_k\|^2/(2\sigma_i^2)]}, \quad p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n}.$$

$$q_{ij} = \frac{(1 + \|z_i - z_j\|^2)^{-1}}{\sum_{a\neq b}(1 + \|z_a - z_b\|^2)^{-1}}, \qquad \min_Z \sum_{i\neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}.$$

- $p$: high-dimensional similarity; $q$: low-dimensional similarity.
- Set $p_{ii} = q_{ii} = 0$; probabilities describe different points. Large $p_{ij}$ but tiny $q_{ij}$ is costly.
- Perplexity controls effective neighborhood size; it is not the number of classes.

# Calculate two conditional neighbor probabilities

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## One point and two other points

Distances are 1 and 2; set $\sigma_i = 1$ and ignore other candidates for this calculation.

$$a = e^{-1/2} \approx 0.607, \quad b = e^{-2} \approx 0.135, \quad p_{1|i} = \frac{a}{a+b} \approx 0.818, \quad p_{2|i} \approx 0.182.$$

- These two conditional probabilities sum to one over the specified candidates.
- Doubling distance from 1 to 2 does not merely halve probability; the squared-distance exponential matters.
- With $\sigma_i = 2$, the distribution becomes flatter, approximately $(0.593, 0.407)$.
- t-SNE then symmetrizes conditional probabilities and compares them with low-dimensional similarities.

# The crowding problem explains the heavy-tailed low-dimensional similarity

- Many moderately distant neighbors in high dimensions cannot all fit at comparable distances around one point in 2-D.
- A heavy-tailed similarity allows dissimilar points to sit farther apart without requiring every local group to collapse.
- Perplexity changes the neighborhood scale; different scales may reveal different parts of the structure.
- Optimization is nonconvex: repeat seeds and compare stable local patterns.
- Cluster sizes, gaps and between-cluster distances in t-SNE are not direct measurements of original density or semantic difference.

**Interpretation**

A visually isolated group is a hypothesis to check, not a discovered class by itself.

# Perplexity controls neighborhood scale, not the number of groups

$$\mathrm{Perp}(P_i) = 2^{H(P_i)}, \qquad H(P_i) = -\sum_j p_{j|i} \log_2 p_{j|i}.$$

- A uniform distribution over four candidates has entropy 2 bits and perplexity 4.
- A distribution concentrated on one candidate has entropy near zero and perplexity near 1.
- Fixing perplexity adjusts the local bandwidth to obtain a target effective neighborhood size.
- Changing this scale can change visible groups; it does not specify their count.
- Compare several reasonable values and inspect stable neighborhoods before interpreting a visual gap.

# Read a t-SNE digit map cautiously




- Colors identify known digit labels for interpretation; t-SNE does not need these labels to fit.
- Nearby points suggest similar local representations; inspect the actual digit images to explain exceptions.
- The gap between groups of zeros and ones is not a calibrated semantic distance.
- An outlying point may be ambiguous writing, noise or an embedding artifact.

**Next action**

Return to original samples and compare neighborhoods across settings.

# A separated display must be checked against original observations

**A point inside the wrong-colored group**

A digit labeled 4 appears near several 9s in a t-SNE display.

1. Inspect its original strokes: is it ambiguous, noisy or mislabeled?
2. Compare its nearest neighbors in the original fitted representation.
3. Repeat nearby settings or seeds to see whether the placement is stable.
4. Use an independent labeled validation task if classification performance is the objective.

- A visual exception is a useful investigation lead, not immediate permission to change the label.
- The plotting coordinates are not a deployed classifier by themselves.

# UMAP also starts from neighborhoods, with a different objective

| Aspect | t-SNE | UMAP |
|---|---|---|
| High-dimensional relation | Neighbor-similarity probabilities | Weighted local neighborhood graph |
| Low-dimensional fit | KL divergence using heavy-tailed similarity | Attraction and repulsion matching graph membership |
| Main controls | Perplexity, learning rate, initialization | Number of neighbors, minimum distance, metric |
| Interpretation | Mostly local exploratory structure | Local structure with setting-dependent broader organization |

- Neither method guarantees that every plotted distance or cluster area has an original-space meaning.
- UMAP commonly provides a fitted transform for new points; compare the actual implementation and use case.

# UMAP combines directed neighbor memberships into a symmetric weight

$$p_{ij} = p_{j|i} + p_{i|j} - p_{j|i}p_{i|j}.$$

## Two directed memberships

If $p_{j|i} = 0.8$ and $p_{i|j} = 0.4$, the symmetric value is $0.8 + 0.4 - 0.32 = 0.88$.

- This has the form of a fuzzy union of the two directed relationships.
- It differs from taking their average, which would give 0.6.
- The resulting graph is matched by attractive and repulsive low-dimensional relationships.
- Neighbor count and minimum-distance settings still shape the displayed arrangement.
- A high graph membership is not a verified probability that two observations share a class.

# Check false neighbors and missing neighbors separately

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## One point, two nearest neighbors

Original neighbors are $\{A, B\}$; embedded neighbors are $\{A, C\}$.

- $C$ is a **false neighbor**: the map pulled an originally distant point close. Trustworthiness penalizes such intrusions.
- $B$ is a **missing neighbor**: the map separated an originally close point. Continuity penalizes such omissions.
- Both scores depend on the neighborhood size and rank severity, not simply on whether clusters look clean.
- For a prediction task, additionally compare a downstream model on held-out labeled data.
- Inspect representative originals, unstable points and several seeds or parameter choices.

# Neighbor overlap exposes a distortion hidden by an attractive plot

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## Three original neighbors

Original set is $\{A, B, C\}$; embedded set is $\{A, C, D\}$.

$$\text{retained-neighbor fraction} = \frac{|\{A, B, C\} \cap \{A, C, D\}|}{3} = \frac{2}{3}.$$

- B is missing and D is an intrusion; the display preserves only two of the three selected relations.
- An overlap fraction ignores ranks. Trustworthiness and continuity additionally penalize how severe the ranking changes are.
- Evaluate at several neighborhood sizes because local quality can differ by scale.
- A coherent-looking cluster layout can still have poor point-level neighborhoods.
- Combine numeric checks with representative original samples.

# Choose a method by the relation you need

| Need | First candidate | Evidence to check |
|---|---|---|
| Linear compression | PCA by SVD | Variance, reconstruction, task score |
| Map from distances | Classical / metric MDS | Euclidean fit; distance residuals |
| Unfold a curved surface | Isomap | Connectivity, shortcuts, path stability |
| Local interpolation | LLE | Stable weights; neighbor coverage |
| Smooth graph coordinates | Laplacian eigenmaps | Relevant edges; nonconstant modes |
| Explore local groups | t-SNE / UMAP | Neighbors, seeds, parameters, originals |

### Principle

The objective determines what the low-dimensional coordinates mean.

# Exit exercise: defend a sensor representation

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## A concrete case

There are 500 sensor features, 2,000 training observations and 300 labeled validation observations. Many features are correlated; the goal is fault classification and a 2-D exploratory display.

1. Propose separate representations for the classifier and for the display.
2. Explain where centering, scaling and PCA are fitted.
3. Name one numerical check and one original-data inspection.
4. Explain why a separated 2-D cluster is not sufficient proof of a fault category.

## What a good answer includes

Task, retained relation, fitted transformation, validation evidence and an explicit limit of the picture.